ELENA RASSOKHINA

PROFILE

Senior full-stack engineer with more than 10 years of experience in IT and database architecture, including over 5.5 years focused on commercial development of distributed cloud applications, microservices, and web interfaces with Node.js (TypeScript) and React.

- **Backend expertise:** Deep knowledge of Node.js (NestJS, Express.js), asynchronous Python (FastAPI), microservice architecture, and BFF and API Gateway patterns.
- **Frontend expertise:** Design of complex SPAs, interactive dashboards, and real-time interfaces with React (TypeScript, Redux Toolkit), following the FSD methodology.
- **Data and high-load systems:** Confident work with very large databases (180+ GB, 20+ million records), optimization of complex SQL queries, and experience with NoSQL (MongoDB), caching (Redis), and messaging (RabbitMQ, Kafka).
- **AI/LLM engineering:** Hands-on experience deploying local AI environments (Ollama, Gemma) and designing end-to-end RAG pipelines.

Languages: Russian (native), English (B2 - fluent technical communication).

TECHNICAL SKILLS

- **Languages and frameworks:** TypeScript, JavaScript (ES6-ES13), Node.js, NestJS, Express.js, React, Redux Toolkit, Python, FastAPI, SQLAlchemy, PHP
- **Databases and caching:** PostgreSQL, MySQL, MSSQL (T-SQL), MongoDB, Redis, InfluxDB, Neo4j, SAP HANA.
- **Message brokers and protocols:** RabbitMQ, Kafka, WebSockets (Socket.io), MQTT, REST API, OData, SOAP, XML.
- **Infrastructure and DevOps: Docker, Docker Compose, Kubernetes (stateless application architecture), GitLab CI/CD, Jenkins, GitHub Actions, Linux (SSH, bash), Nginx.**
- **Testing and monitoring:** Jest, Chai, Mocha, PyTest, Apache JMeter, Grafana, Kibana, Graylog, SonarQube.
- **AI and LLM stack: Ollama, Gemma, RAG pipelines, intelligent search and embedding systems, Cursor.**

PROFESSIONAL EXPERIENCE

Senior Software Engineer at Creative Way, Moscow

March 2025 - Present

Technology stack: Python (FastAPI, asyncio, SQLAlchemy, Alembic), Node.js (NestJS, Express.js), React (TypeScript, Redux Toolkit), PostgreSQL, InfluxDB, Docker, RabbitMQ, WebSockets, MQTT, OPC, Ollama, Gemma, RAG.

Responsibilities:

- **Design and development of AI/LLM environments:** designed and implemented local architectural solutions based on Ollama and Gemma to automate the analysis of corporate regulatory documentation.
- **Distributed systems development:** designed microservice backend logic in Python (FastAPI) and Node.js (NestJS), configured REST API contracts, and built secure end-to-end authorization services (SSO/JWT).
- **IoT telemetry and Big Data integration:** optimized data layers and configured high-throughput asynchronous clients for collecting and visualizing real-time data.
- **Infrastructure ownership:** independently managed full-stack delivery in cross-functional Scrum teams of up to 10 people, covering containerization (Docker), CI/CD pipeline automation, and production incident analysis.

Achievements:

- **Built a RAG pipeline from scratch:** designed semantic document chunking, embedding index construction, conversational search, and generation of draft regulations with automatic source citations.
- **Implemented a fault-tolerant notification service:** delivered event processing and asynchronous routing through RabbitMQ, state storage in PostgreSQL/Sequelize, and sub-second delivery to the React client via WebSockets.
- **Integrated a real-time location system (RTLS):** developed an MQTT client to collect and process employee positioning data, followed by real-time dashboard visualization, improving safety at a major airline's production facility.
- **Optimized industrial telemetry collection:** wrote an asynchronous client for collecting data from plant OPC servers (SCADA) and organized scalable time-series storage in InfluxDB for predictive analytics.
- **Skolkovo resident status:** prepared technical documentation and product architecture diagrams, enabling the product to pass the audit required to obtain Skolkovo resident status.

Senior Software Engineer at EPAM Systems Latvia, Riga

September 2023 - March 2025

Technology stack: JavaScript (ES13), Node.js (Express.js, SAP CAP), TypeScript, Jest, Chai, OData, SAP BTP (Cloud Foundry), SAP Fiori (UI5), SAP HANA (SQL), Azure AD, Kibana, SonarQube, Git, Jira.

Responsibilities:

- **Data architecture design:** designed complex relational database structures (entity-relationship diagrams) and developed enterprise-grade data storage schemas.
- **Cloud API development:** developed and maintained scalable REST/OData APIs for Node.js microservices, and configured and administered environments in cloud infrastructure (Cloud Foundry and PaaS solutions).
- **Enterprise systems integration:** implemented secure cross-tenant access to internal resources and integrated corporate authorization services.
- **Frontend development:** created the client side of business applications in JavaScript (ES13) based on stakeholder requirements.

Achievements:

- **Standardized enterprise integration:** developed and implemented service business logic in Node.js using OData, accelerating the delivery of new client requirements to production by 30%.
- **Automated enterprise authorization:** integrated internal cloud services with Microsoft Entra B2B / Azure AD and implemented end-to-end Single Sign-On (SSO) authentication for seamless onboarding of external clients.
- **Designed data structures:** developed ERD schemas for major international clients (a German sportswear brand and an American online travel agency), ensuring database scalability throughout the product lifecycle.
- **Optimized background processes:** configured a Node.js job scheduler and implemented automated asynchronous email notification delivery.

Senior Software Engineer at Accenture Baltics, Riga

February 2021 - September 2023

Technology stack: JavaScript (ES12), Node.js (Express.js, SAP CAP), Jest, Chai, Mocha, Postman, Apache JMeter, AWS S3, Jenkins, Groovy, Kibana, Grafana, SonarQube, Git, Jira.

Responsibilities:

- **Core logic design and development:** designed data models (ERD) and developed REST API services in Node.js for monolithic and microservice cloud architectures.
- **CI/CD and QA automation:** configured automated application build, testing, and deployment pipelines in cloud infrastructure, and wrote automation scripts.
- **Load testing and profiling:** identified performance bottlenecks in distributed systems and configured end-to-end monitoring.
- **Mentoring:** served as a technical mentor for junior developers, conducted code reviews, and introduced quality standards.

Achievements:

- **Developed a comprehensive ERP/SaaS solution:** designed and launched modular integration for B2B services (invoicing, billing, and order management) in Node.js with secure JWT- and OAuth 2.0-based authorization.
- **Automated CI/CD pipelines:** wrote automated Groovy scripts for Jenkins CI/CD, reducing manual effort and shortening the release cycle.
- **Implemented load testing:** developed a series of stress tests in Apache JMeter, configured systematic pipeline runs, and visualized distributed-system performance metrics in Grafana.
- **Implemented a geo-distributed service:** optimized algorithms for filtering and serving promotions by geographic location in Node.js, enabling major international clients from Italy and Germany to adapt content instantly for local markets.
- **Optimized infrastructure costs:** integrated the system's file storage with AWS S3, reducing load on core application servers and improving media-storage resilience.

Software Engineer (PHP / Full-Stack / Database Developer) at the Agricultural Data Center, Riga

October 2012 - February 2021

Technology stack: T-SQL (MSSQL), PHP 8.0, JavaScript, jQuery, Bootstrap, HTML5, CSS3, REST API, SOAP, XML, Git.

Role overview:

Designed, optimized, and maintained very large government relational databases (20+ million records, 180+ GB). Wrote and refactored complex server-side database logic, including triggers, views, and stored procedures in T-SQL. Created internal administration panels, user portals, and accounting modules in PHP and JavaScript (jQuery).

Achievements:

- **Improved core database performance several-fold:** conducted a full database-structure audit, rewrote expensive T-SQL queries, and configured indexes, eliminating critical delays in analytical report generation.

- **Integrated the system with a pan-European platform:** developed and implemented secure integration gateways (REST/SOAP API) to synchronize data with TRACES (Trade Control and Expert System), the European system for monitoring the movement of goods.
- **Migrated legacy code:** successfully moved key internal ministry web services from older PHP versions to modern object-oriented PHP 8.0 without downtime for the government system.